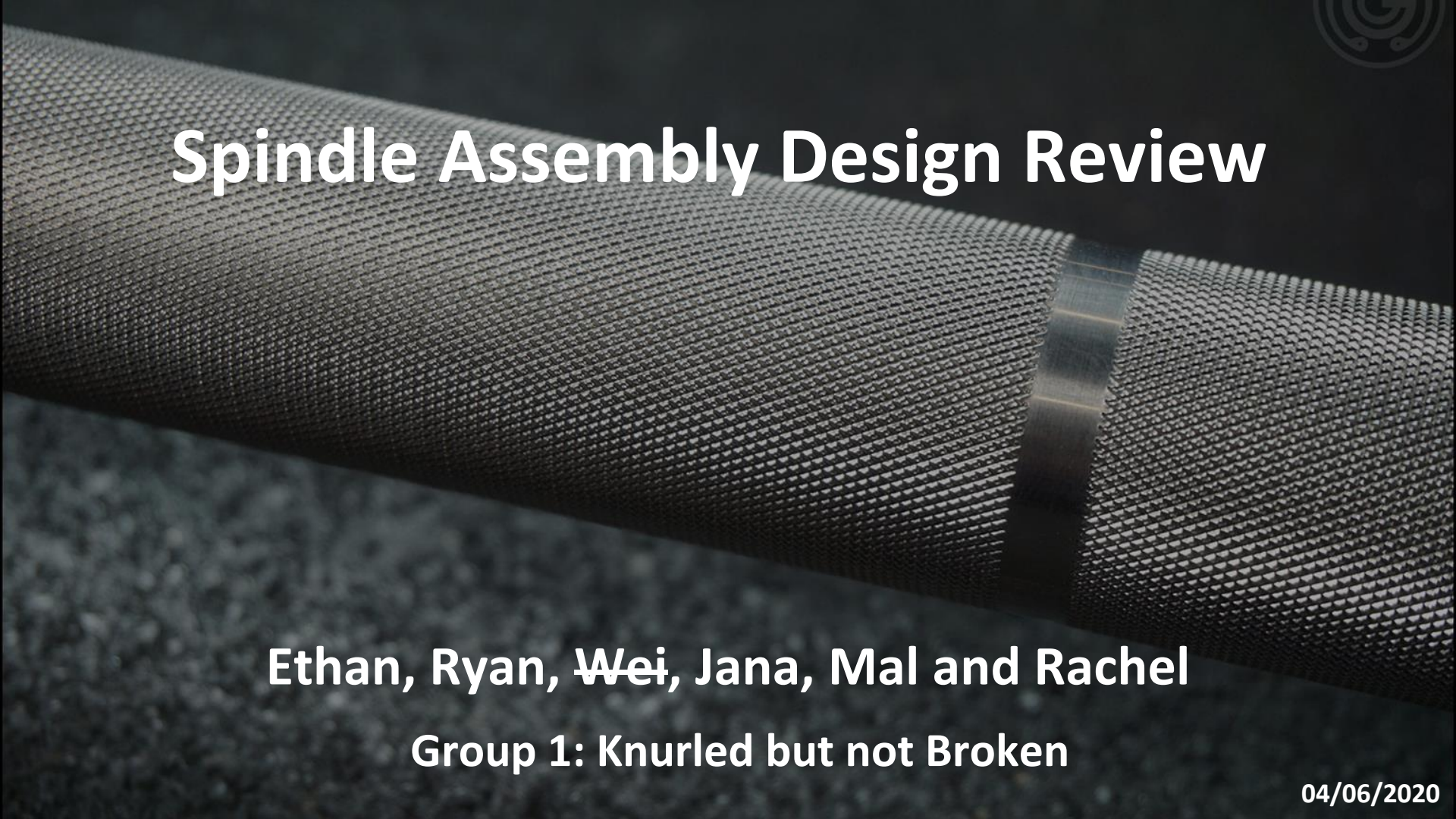


A close-up photograph of a cylindrical metal spindle assembly. The surface is covered in a fine, uniform knurled pattern. A dark, rectangular band or tape is wrapped around the middle of the spindle. The lighting is dramatic, highlighting the texture of the knurling and the band.

Spindle Assembly Design Review

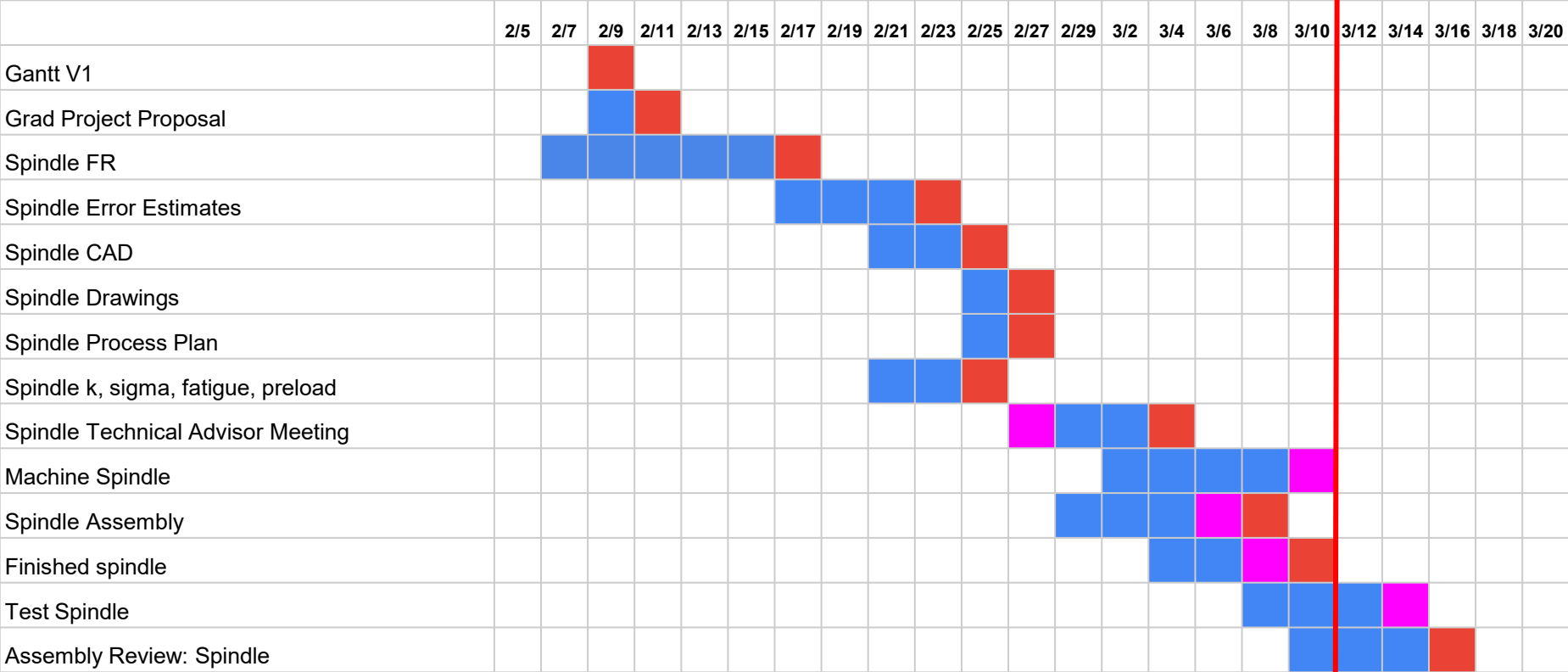
Ethan, Ryan, Wei, Jana, Mal and Rachel

Group 1: Knurled but not Broken

04/06/2020

Gantt Chart-Spindle

On Schedule



System Level FRs

Parameter	Value	Units	Justification	References
Material to cut	12L14 Steel		Staff Requirement	Requirement Sheet
MRR	0.08 (min)	in ³ /min	Staff Requirement	Requirement Sheet
Surface Finish	63 (max)	μin	Staff Requirement	Requirement Sheet
Repeatability	50 (max)	μm	Staff Requirement	Requirement Sheet
Drop impulse force	2000	N	Staff Requirement	Drop impulse force 3-5x gravity load
Weight on carriage	600	N	Staff Requirement	Weight on carriage of average sized team member
Sledgehammer force	5000	N	Staff Requirement	Sledgehammer force $\frac{1}{2}mv^2$

System Level FRs

Parameter	Value	Units	Justification	References
Maximum workpiece Diameter	15 (min)	mm	Limit of chuck size (internal jaws)	Chuck Datasheet
Machine Life	2000 (min)	hours	Desired machine life for hobbyist or mini shop	1 work year
Maximum runout at end of spindle shaft	50 (max)	μm	Typical values for lathes	*Measure on existing lathe (future work)

Spindle FRs

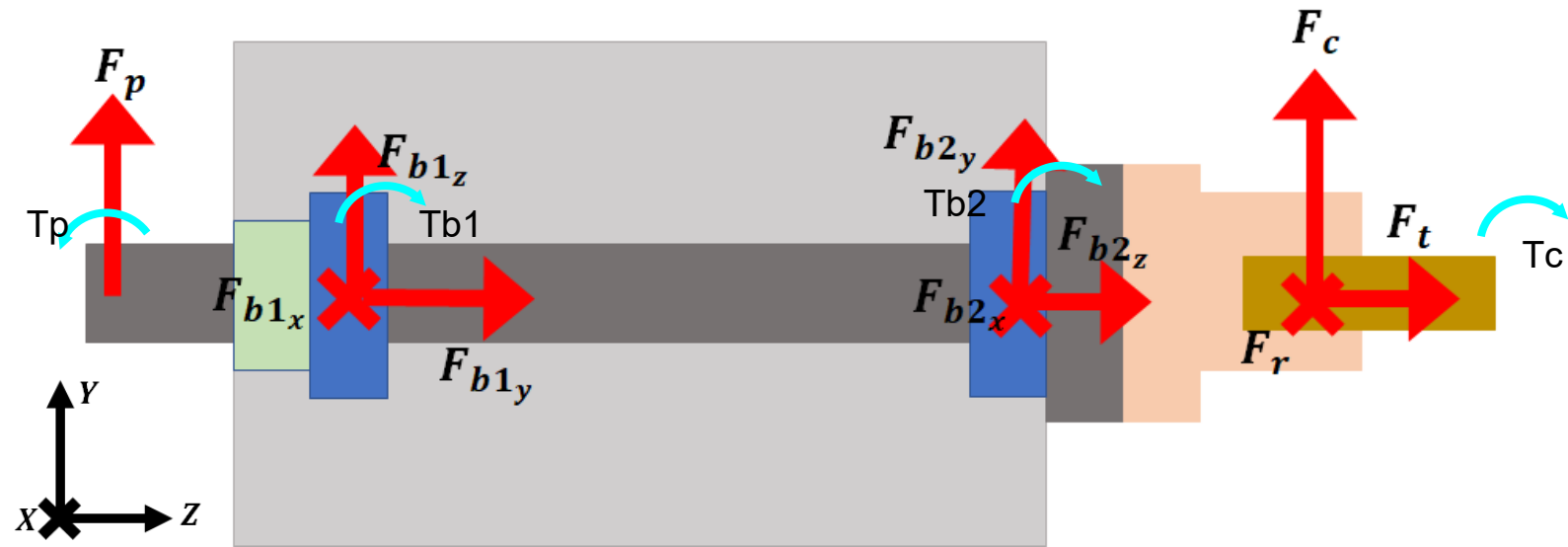
Parameter	Value	Units	Justification	References
Dissipated power	100 (max)	W	$u_t = 9 \text{ J/mm}^3$ MRR = $21.85 \text{ mm}^3/\text{s}$ Cutting power = 200 W Max motor power = 337 W	Machinery's Handbook Kalpakjian Motor spec sheet
Net Axial Stiffness	35×10^6 (min)	N/m	Max thrust force = 400 N Allowable deflection = $11.3 \mu\text{m}$	
Net Radial Stiffness	17×10^6 (min)	N/m	Max cutting force = 200 N Allowable deflection = $11.3 \mu\text{m}$	
Machine Life	2000 (min)	Hours	1 work year (8 hrs/day, 5 days/week, 52 weeks/year)	
Temperature of components	40 (max)	Deg C	Thermal imaging experiments using past years' lathes	

Summary of Requirements and Design

Parameter	Value	Units	Design Choices	Measurement to verify
Dissipated power	100 (max)	W	<u>Grease</u> : Timken GR236 <u>Transmission ratio</u> : 1:2 or less	Measure motor operating point (multimeter, current sensor, etc.)
Net Axial Stiffness	35×10^6 (min)	N/m	<u>Preload</u> : 400N	Clamp down spindle, Load with known force, measure deflection with dial indicator
Net Radial Stiffness	17×10^6 (min)	N/m	<u>Bearing Spacing</u> : 2.6 in <u>Preload</u> : 400 N	Clamp down spindle, Load with known force, measure deflection with dial indicator
Machine Life	2000 (min)	Hours	<u>Bearing Grease</u> : Timken GR236 <u>Preload</u> : 400 N	Run to failure
Temperature of components	40 (max)	Deg C	<u>Grease</u> : Timken GR236	Use thermal imaging

Free Body Diagram

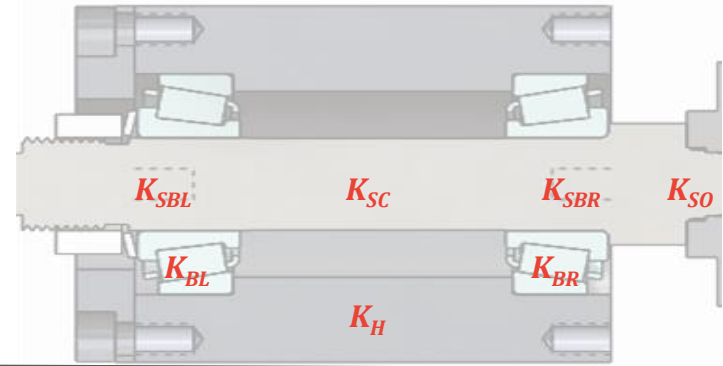
Fb	Bearing Reaction Forces	Fr	Radial Force from Cutting
Fp	Pulley Forces	Tp	Torque from Pulley
Fc	Cutting Force	Tb	Torque from Bearings
Ft	Thrust Force from Cutting	Tc	Torque from Cutting Force



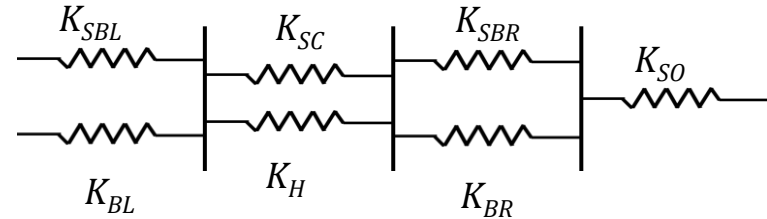
Spindle Stiffness Model

K-axial-total	$3.7 \times 10^8 \text{ N/m}$
K-radial-total	$5.6 \times 10^7 \text{ N/m}$
Ksr / K-radial-total	.0352
Kbr / K-radial-total	.578
Khr / K-radial-total	9.16
Ksa / K-axial-total	1.21
Kba / K-axial-total	0.014
Kha / K-axial-total	6.09

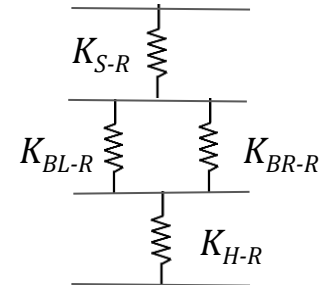
Bearings drive **axial stiffness**, Shaft drives **radial stiffness**



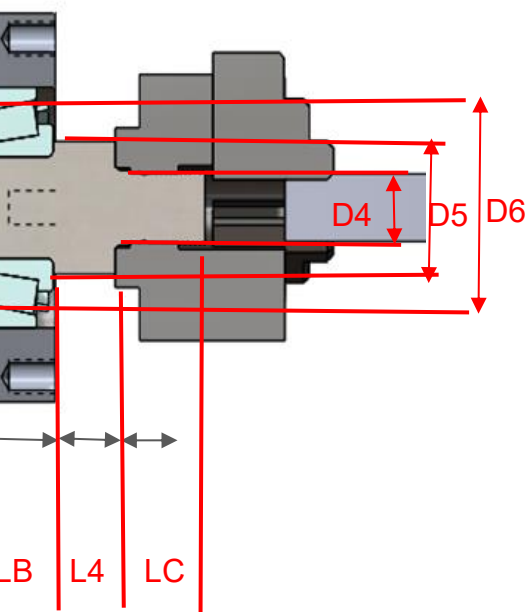
Axial Compliance Model



Radial/Lateral Compliance Model

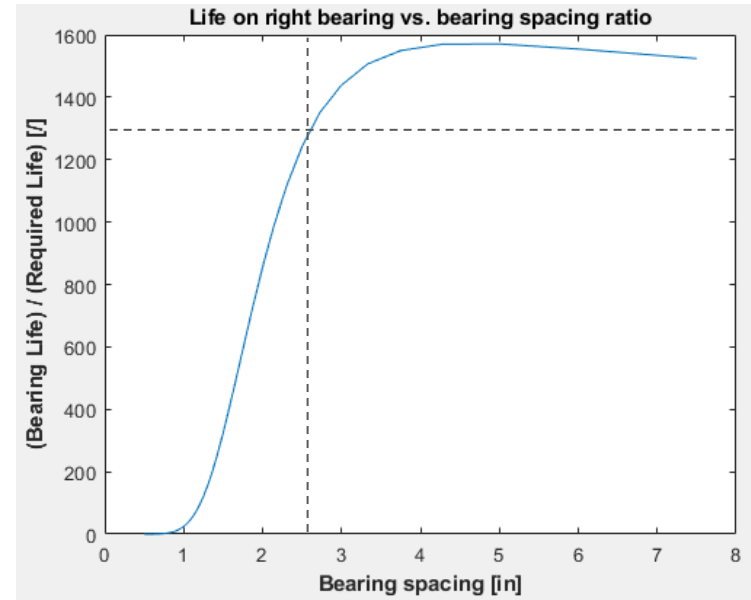
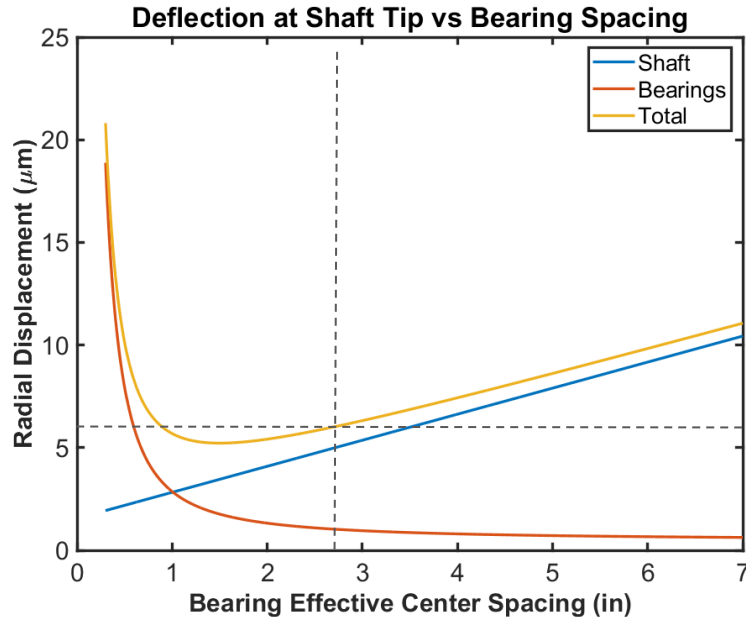
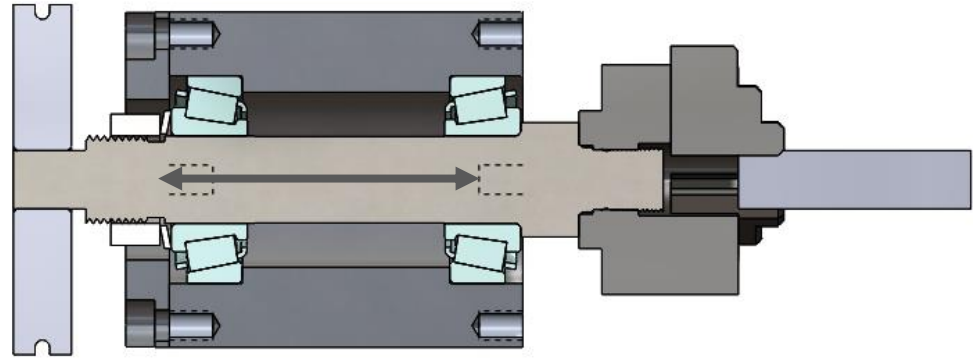


Spindle Geometry

Legend		Dim	Obtained From
Dimensions Provided	Dimensions We Designed	D1	Pulley Bore
	D2	Bearing ID	
	D3	Bearing OD	
	D4	Chuck Bore	
	D5	Bearing Specs	
	D6	Bearing Specs	
	L1	Pulley/gear width	
	L2	Preload spacing	
	L3	Bearing Stiffness/Life	
	L4	Shoulder loading/bending	
	LB	Bearing width spec	
	LC	Chuck bore length	

Spindle Geometry

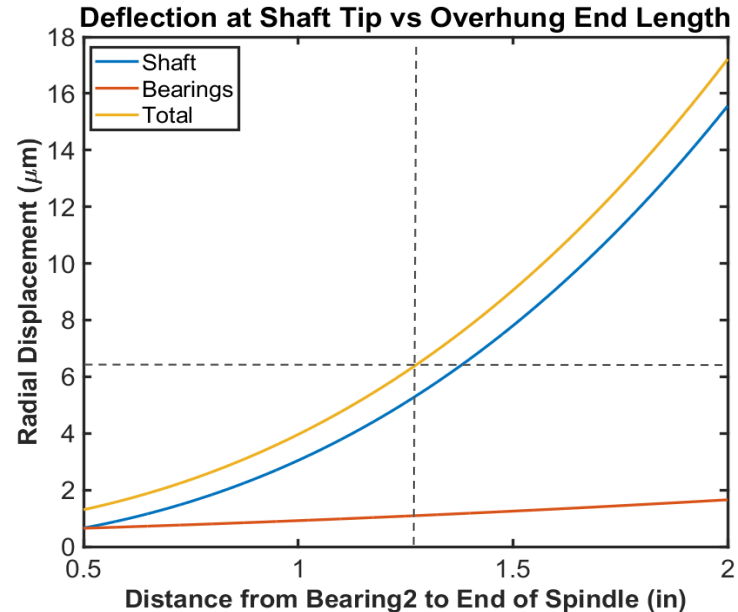
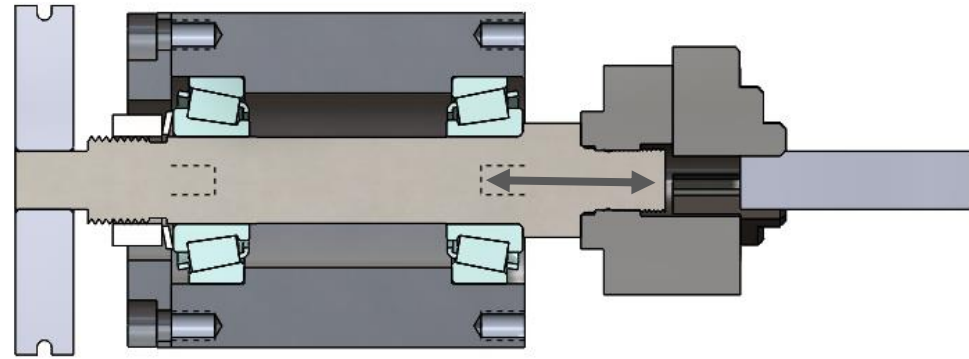
Decreasing spacing between bearings and decreasing dist between bearing 2 and chuck will **decrease deflection** at tip. Increasing bearing spacing extends **bearing life** with little impact on **shaft deflection**



Spindle Geometry: Effect on Bearing Life

Decreasing dist between bearing 2 and chuck will **decrease deflection** at tip. However, the head stock's thickness would need to be reduced or the housing would need to be mounted differently

Due to Abbe Errors, thinner head stock portions will allow more deflection (~10 microns)

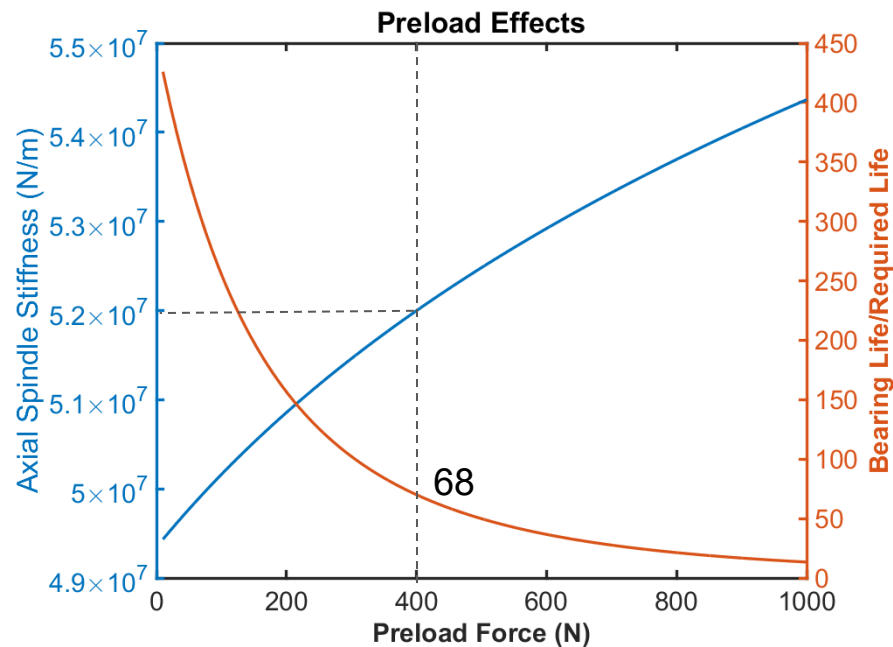
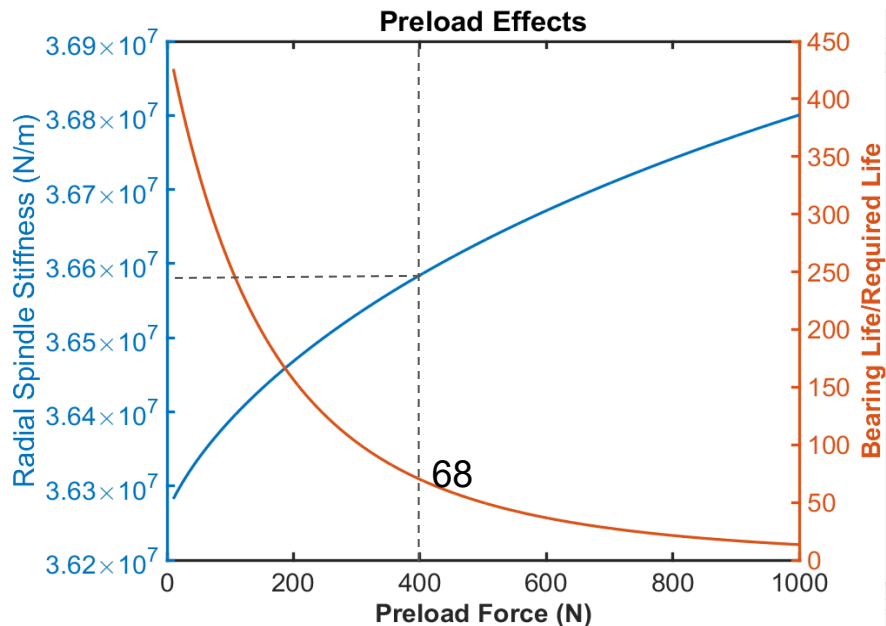


Bearing Preload

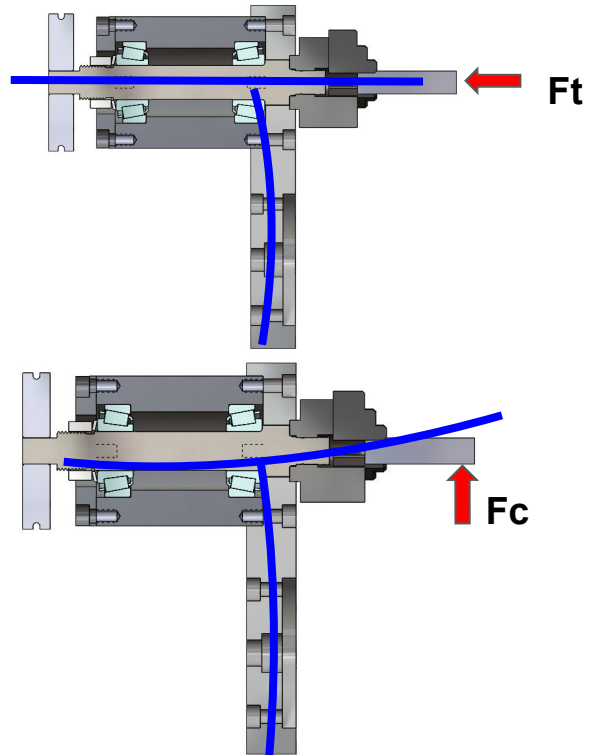
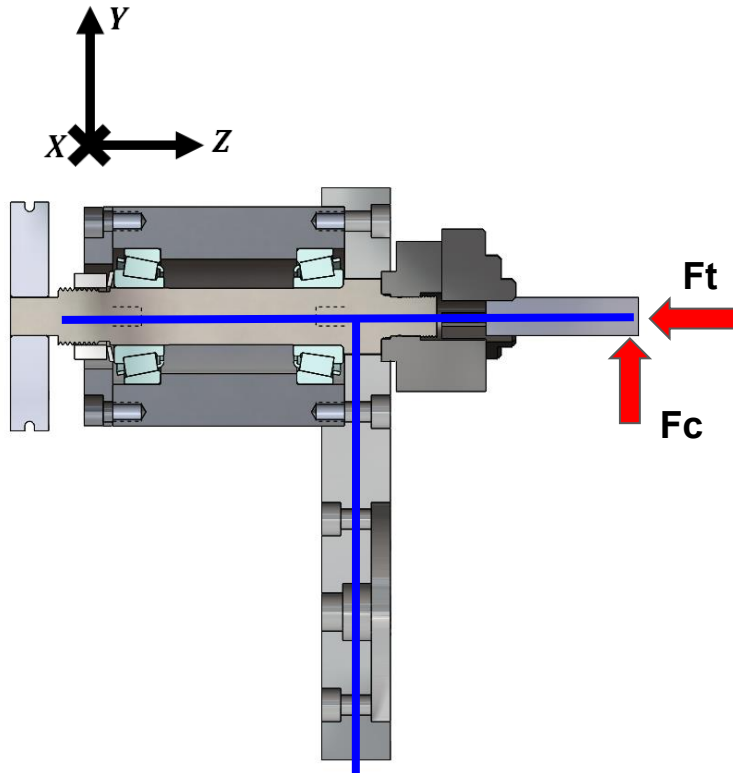
The preload sets the **stiffness** of the spindle by controlling the fit between the cup and cone of the bearings. Too much preload will increase forces on the bearing and decrease the **life** of the spindle.

$$K_{\text{Spindle}} = \left[\frac{1}{K_{\text{shaft}}} + \frac{1}{K_{\text{bearing}}} \right]^{-1}$$

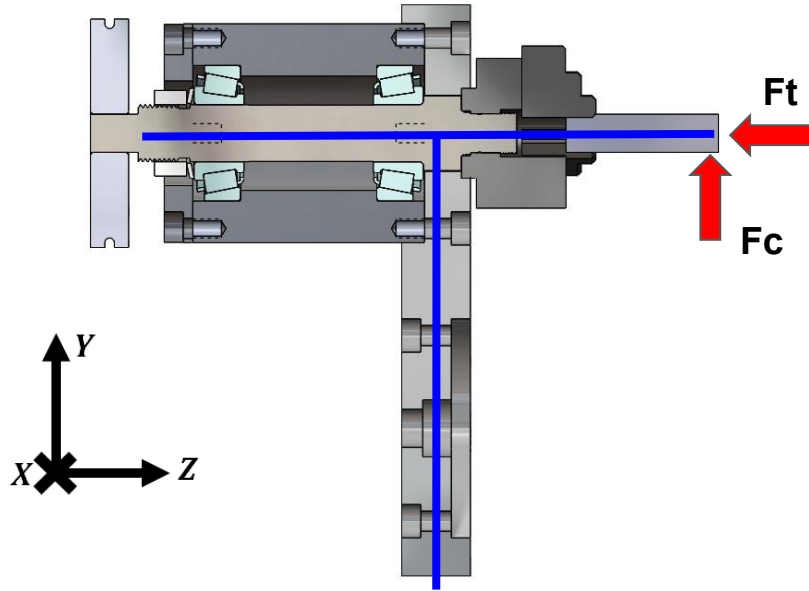
From FR: Bearing Life Min = 2000 hours, axial K = 3.5×10^7 N/m, radial K = 1.7×10^7 N/m



System Error Budget



System Error Budget



F_t	$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -4.03 \\ -11.06 \end{bmatrix} \mu m$
F_c	$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 12.04 \\ -6.27 \end{bmatrix} \mu m$
Thermal	$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2.9 \end{bmatrix} \mu m$
Total	$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 8.01 \\ -20.23 \end{bmatrix} \mu m$

RSS= 21.7 μm

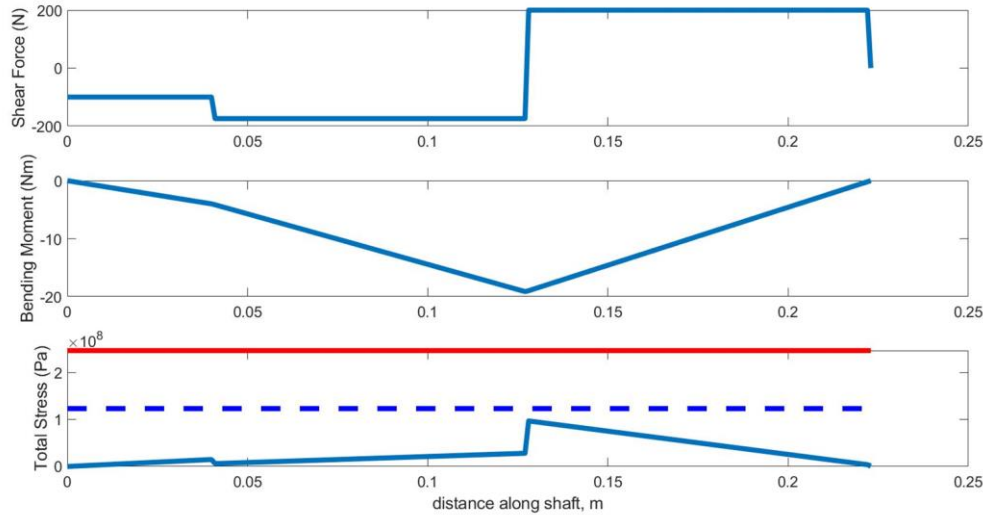
Runout must be measured and included in estimate

Effective Stiffness Axial = 1.73E7 N/m

Effective Stiffness Radial = 2.49e+07 N/m

So far, we meet our desired spec of 28.86 μm allowable error per axis to meet a total of 50 μm RSS repeatability error.

Fatigue and Spindle Shaft Life

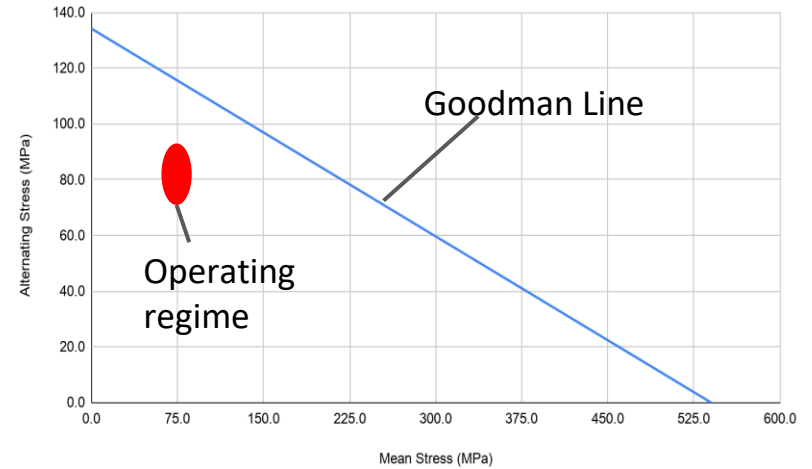


Shaft Stress 

Fatigue Stress 

Yield Stress 

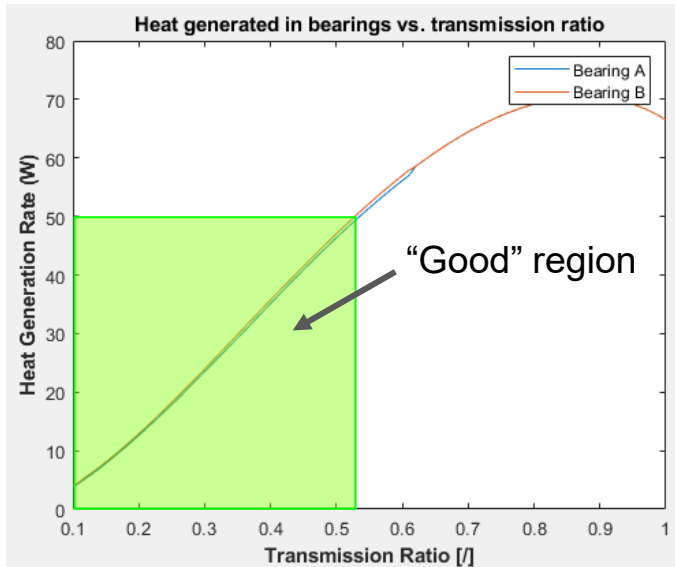
Mean Stress v. Alternating Stress



Cutting loads that meet material removal requirements
allow for operation below the Goodman Line

Transmission Ratio

- Primary decision factor: **Heat Generation in Bearings**
- **↑ Transmission Ratio == ↑ Heat Generation**



If transmission ratio is 0.52 or less, heat dissipation FR of 50 W/bearing is met

Transmission ratio =

(pulley diameter on motor) / (pulley diameter on spindle shaft)

Bearing Grease

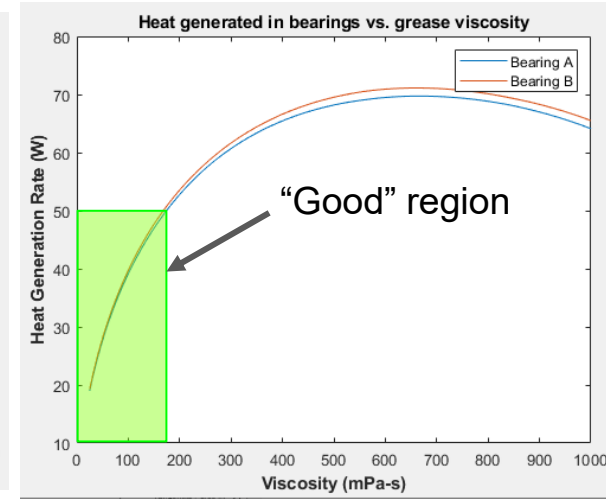
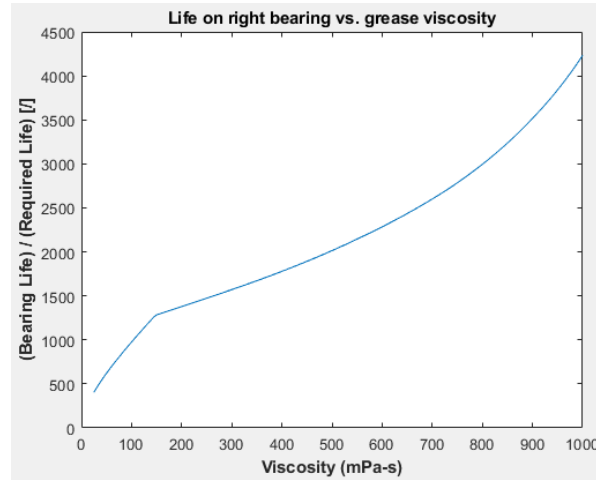
Should pick viscosity so that:

- Heat dissipation < 50 W/bearing
- Acceptable life
- Lowest cost

↓ Grease viscosity

- ↓ Bearing heat
- ↓ Bearing life

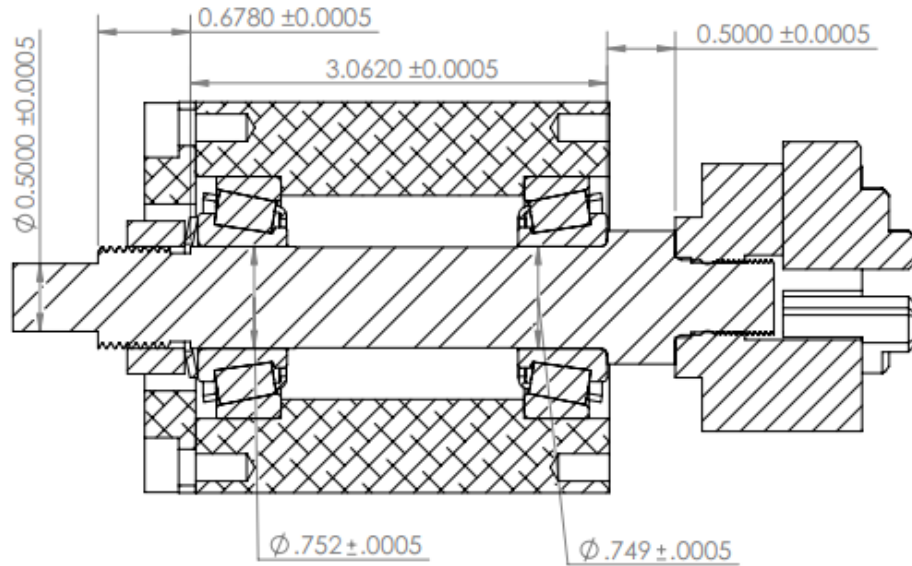
Viscosity of Timken GR236 grease meets heat dissipation requirement with lowest cost



Manufacturer	Product Number	Viscosity (cSt)	Cost (\$)
Timken	GR220	120	9.93
Timken	GR217	240	7.20
Timken	GR232	506	22.34
Timken	GR219	490	11.45
Timken	GR231	110	15.90
Timken	GR236/237	145.6	5.50

Spindle Manufacturing

Key Dimensions



All dimensions measured to within spec at the time of manufacturing, but do not have actual measurements on hand



We completed our spindle,
but were unable to get data,
photos or measurements





Appendix

Cutting Forces & Power

Assumed Parameter	Value	Units	Justification
Material removal rate (MRR)	0.08	in ³ /min	Specified by staff
Material specific cutting energy	9	J/mm ³	High estimate for steel, Kalpakjian + MH
Tool rake angle	0	rad	Worst case; rake angle ↓, cutting forces ↑
Coefficient of friction between tool & part	2	/	Worst case according to Kalpakjian
Part diameter	15	mm	Max diameter for internal jaws, chuck spec sheet

Maximum Cutting Power: 200W

Maximum Cutting Force: 200N

Maximum Thrust Force: 400 N

Cutting Forces: Equations

Equation	Justification	Reference
$V_c = \pi D N / 60$		$V = (\text{Radius}) * (\text{rot. speed})$ N in RPM
$F_c = (u_t * \text{MRR}) / V_c$		Conservative estimate; all power goes into cutting
$\tan(\beta) = \mu$		Kalpakjian
$F_t = F_c * \tan(\beta - \alpha)$		Kalpakjian
Friction torques from bearings (MA and MB)		Timken manual
$T_m = (F_c * (D/2) + MA + MB) * n$	FBD on spindle shaft	n = transmission ratio from motor shaft to spindle shaft
$N_{\text{new}} = \omega_{nl} * (1 - (T_m / T_{\text{stall}})) * n$	Torque-speed relation for DC motor	ω_{nl} = no load speed of motor T_{stall} = stall torque of motor Values given in motor spec sheet

Continue to iterate multiple times to allow solution to converge

Bearing Life Calculations

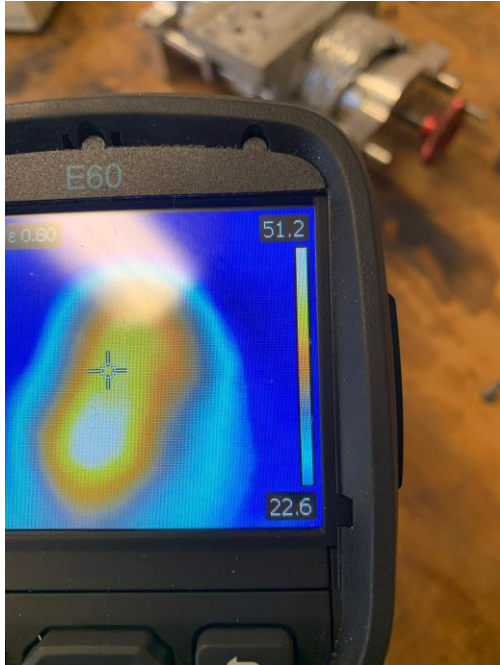
$$L_{na} = a_1 a_2 a_{3d} a_{3k} a_{3l} a_{3m} a_{3p} \left(\frac{C}{P_r} \right)^e (1 \times 10^6) \text{ revolutions}$$

Parameter	Value/calculation	Reference
C = dynamic radial load rating	39100 N	Spec sheet
e	10/3	Pg. 48, for roller bearing
P _r = dynamic equivalent radial load	0.4*F _r + K*F _a	Pg. 43
a ₁ = reliability life factor	fn(reliability)	Pg. 49
a ₂ = material life factor	1.0 for Timken steel	Pg. 49
a _{3d} = debris life factor	1.0	Pg. 49, no other info??
a _{3k} = load zone factor	1.0	Pg. 50, no other info??
a _{3l} = lubrication life factor	= C _g *C _l *C _j *C _s *C _v *C _{gr}	Pg. 50-52
a _{3m} = misalignment life factor	1.0	Pg. 54, no other info??
a _{3p} = low-life factor	1.0 (worst case)	Pg. 54, fn(a _{3l} , C/P _r)

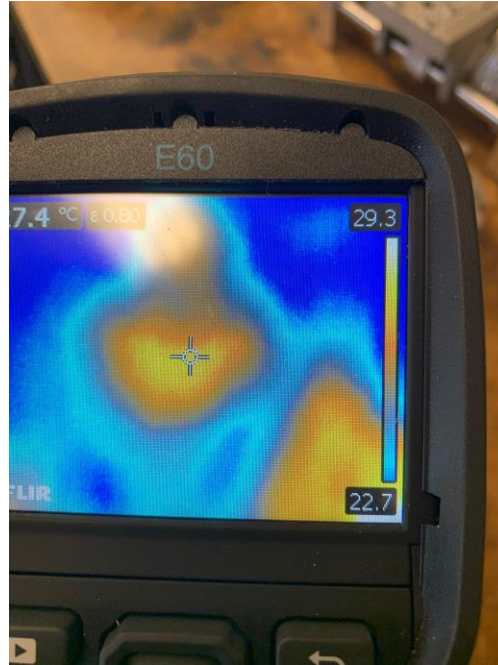
Fatigue

Ideal Endurance Limit, S'_e	216.0	MPa	$S_{ut} / (S'_e/S_{ut})$	
Surface condition modification factor, k_a	0.851		$a * (S_{ut})^b$	Shigley 9th Ed., Equation 6-19
Size modification factor, k_b	0.906		$0.879 * d^{(-0.107)}$	Shigley 9th Ed., Equation 6-20
Load modification factor, k_c	1		$k_c = 1$ for combined torsion + loading	Shigley 9th Ed., Equation 6-26, Footnote 17
Temperature modification factor, k_d	0.99		$0.975 + 0.432(10^{-3})TF - 0.115(10^{-5})TF^2 + 0.104(10^{-8})TF^3 - 0.595(10^{-12})TF^4$	Shigley 9th Ed., Equation 6-27
Reliability factor, k_e	0.814		Table lookup	Shigley 9th Ed., Table 6-5
Actual Endurance Limit, S_e	136.7	MPa	$k_a * k_b * k_c * k_d * k_e * k_f * S'_e$	Shigley 9th Ed., Equation 6-18
Shaft area, A	2.850E-04	m^2	$\pi * (d/2)^2$	
Shaft Area Moment of Inertia, I	6.46E-09	m^4	$\pi/64 * (d^4)$	
Moment from normal force, M	30.480	N-m	$F * l$	
Max Alternating Shaft Stress, σ_a	44.91	MPa	$(M * (d/2))/I$	
Max Alternating Shaft Stress with concentration, $K_t * \sigma_a$	89.82	MPa	$(M * (d/2))/I$	Max Alternating Shaft Stress with concentration, $K_t * \sigma_a$

Thermal Considerations



Motor



Disassembled Bearings

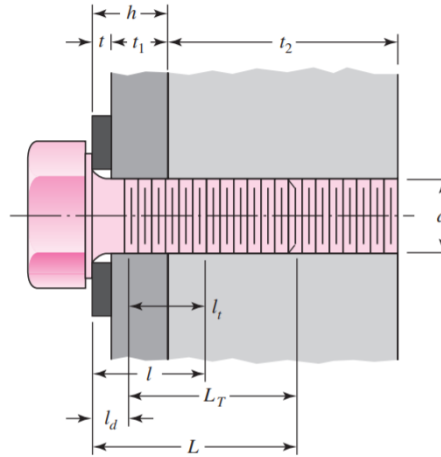


Operating

Bolted Joints

Bolt Stiffness

$$k_b = \frac{A_d A_t E}{A_d l_t + A_t l_d}$$

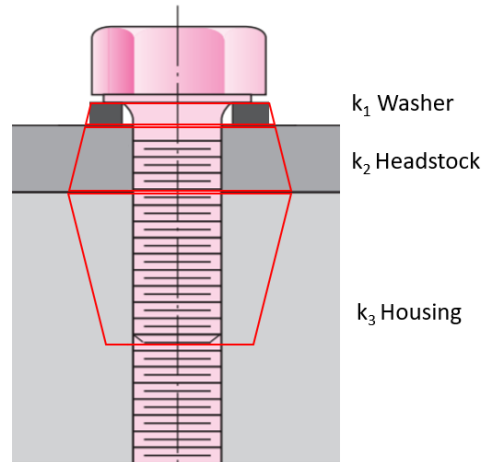


A_d : Area of unthreaded portion

A_t : Area of threaded portion (table 8-2 Shigley)

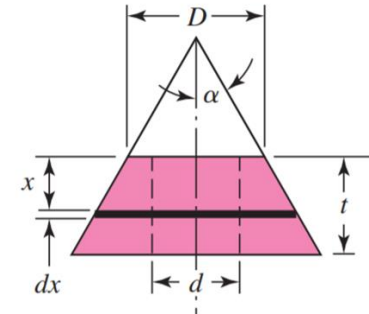
k_b : 1.22 Mlbf/in

k_m : 5.49 Mlbf/in



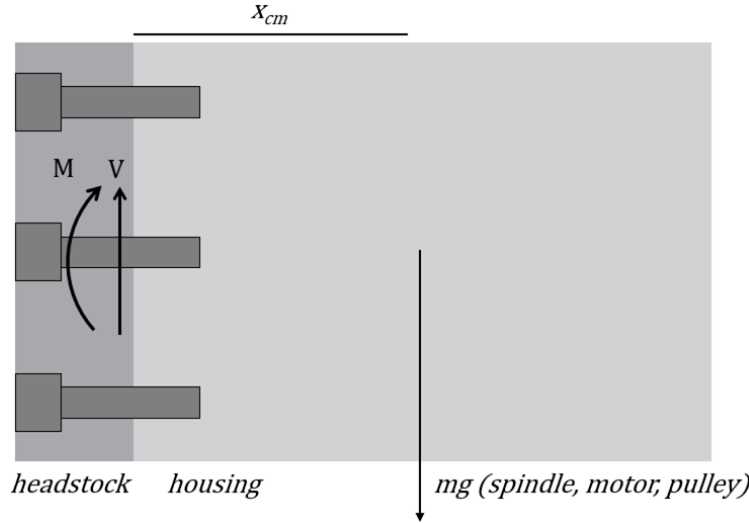
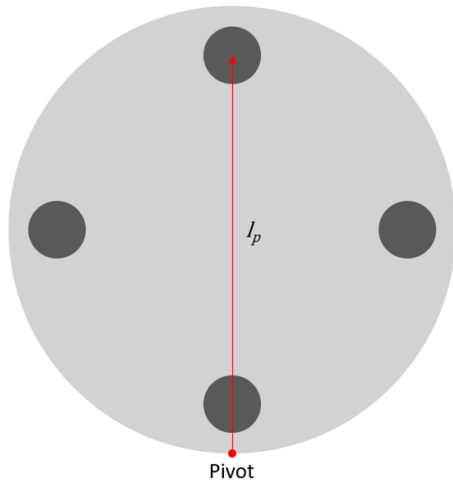
Member Stiffness

$$k = \frac{0.5774\pi E d}{\ln \frac{(1.155t + D - d)(D + d)}{(1.155t + D + d)(D - d)}}$$



$$\frac{1}{k_m} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}$$

Tensile and Shear Stresses in Bolted Joint



Tensile Stress

$$\sigma_b = \frac{CP + F_i}{A_t}$$

$$P = \frac{M}{l_p}$$

Shear Stress

$$\sigma_s = \frac{V}{4A_s}$$

$$\sigma_b : 29.8 \text{ kpsi}$$

$$\sigma_s : 0.1019 \text{ kpsi}$$

$$n_p : 1.11$$

σ_b : Tensile stress in bolt

C: Joint constant

P: Largest external tensile load in bolt

F_i : Preload force

A_t : Tensile stress area (table 8-2)

M: Moment caused by load

l_p : Distance from pivot to top bolt

σ_s : Shear stress in bolt

V: Shear force caused by load and friction between members

A_s : Shear area

n_p : Yielding factor of safety

HTM For Spindle

Actual Coordinates of POI in CS1 due to loads and errors						
	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords
x	35.2950	0.0010	0.0000	0.0015	0.0025	35.2975 mm
y	20.2310	0.0010	0.0000	0.0037	0.0047	20.2357 mm
z	-7.9375	0.0010	0.0000	0.0087	0.0098	-7.9473 mm

Actual Coordinates of CS1 in CS2 due to loads and errors						
	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords
x	-37.1784	0.0010	0.0000	-0.0003	0.0013	-37.1797 mm
y	0.0000	0.0010	0.0000	0.0002	0.0012	0.0000 mm
z	-0.2877	0.0010	0.0000	0.0434	0.0434	-0.3311 mm

Actual Coordinates of CS2 in CS3 due to loads and errors						
	0.0000 Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords
x	74.5490	0.0010	0.0000	-0.0130	0.0140	74.5630 mm
y	12.7000	0.0010	0.0000	-0.0001	0.0012	12.7012 mm
z	0.0000	0.0010	0.0000	0.1103	0.1103	0.0000 mm

Actual Coordinates of CS3 in CS4 due to loads and errors						
	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords
x	-74.5490	0.0010	0.0000	0.0056	0.0066	-74.5556 mm
y	65.4830	0.0010	0.0000	-0.0002	0.0012	65.4842 mm
z	0.0000	0.0010	0.0000	-0.1475	0.1486	0.0000 mm

Actual Coordinates of CS4 in CS5 due to loads and errors						
	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords
x	0.0000	0.0010	0.0000	-0.0002	0.0012	0.0000 mm
y	0.0000	0.0010	0.0000	0.1120	0.1120	0.0000 mm
z	102.3234	0.0010	0.0000	-0.0001	0.0011	102.3245 mm

Actual Coordinates of CS5 in CS6 due to loads and errors						
	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords
x	0.0000	0.0010	0.0000	0.0493	0.0509	0.0000 mm
y	-98.4250	0.0010	0.0000	0.0042	0.0052	-98.4302 mm
z	3.8100	0.0010	0.0000	0.1067	0.1078	3.9178 mm

Actual Coordinates of CS6 in CS7 due to loads and errors						
	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords
x	0.0000	0.0010	0.0000	-0.0168	0.0178	0.0000 mm
y	0.0000	0.0010	0.0000	0.0727	0.0738	0.0000 mm
z	-57.0650	0.0010	0.0000	0.0000	0.0011	-57.0661 mm

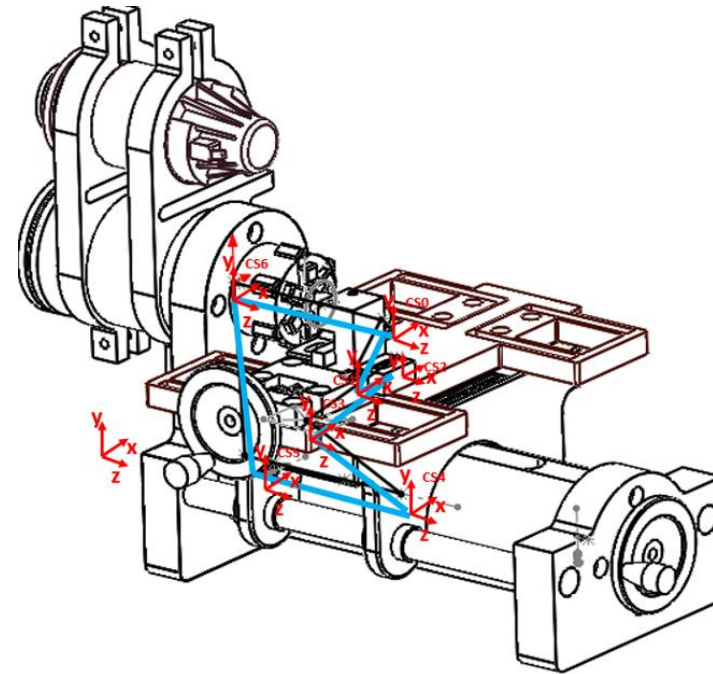
Actual Coordinates of CS7 in CS8 due to loads and errors						
	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords
x	1.8834	0.0010	0.0000	0.0000	0.0010	1.8844 mm
y	-0.0102	0.0010	0.0000	0.0000	0.0010	-0.0112 mm
z	-40.8432	0.0010	0.0000	0.0000	0.0010	-40.8442 mm

Total Error		
dx	0.009685 mm	9.684501 um
dy	0.000828 mm	0.828184 um
dz	0.053585 mm	53.58515 um

Coordinate Summaries			
	X (mm)	Y(mm)	Z(mm)
POI location in CS_1 coordinates	35.295	20.231	-7.938
CS_1 location in CS_2 coordinates	-37.18	0	-0.288
CS_2 location in CS_3 coordinates	74.549	12.7	0
CS_3 location in CS_4 coordinates	-74.549	65.483	0
CS_4 location in CS_5 coordinates	0	0	102.3
CS_5 location in CS_6 coordinates	0	-98.43	3.81
CS_6 location in CS_7 coordinates	0	0	-57.07
CS_7 in CS_poi coords (just for chekcing)	1.8834	-0.01	-40.84
Sum for checking (should be zero)	5.3E-15	-0.021	0

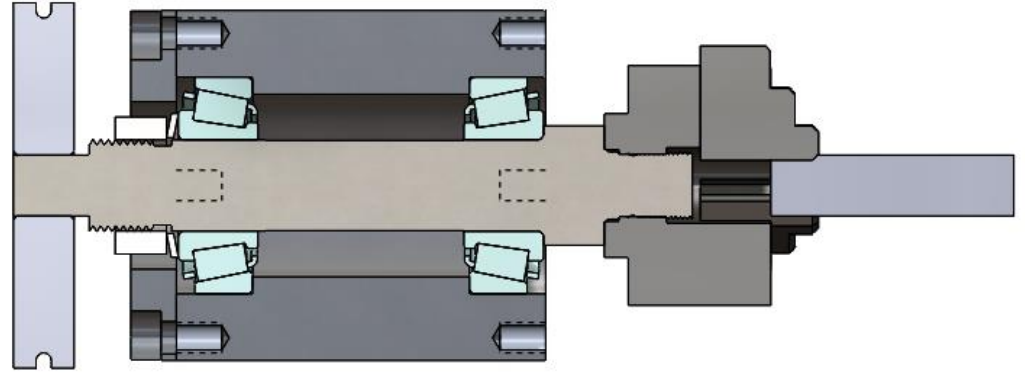
The Z-axis is the most sensitive direction given the current stiffness values. The headstock deflection and the z carriage deflection are two of the biggest causes of this and they would need to be designed with greater care

If every C.S assembly was designed with 1E10 N/m stiffness along every axis (trans. And rot) the system would well be within our error budget.



Spindle Geometry

Choosing dist between bearing 2 and chuck



By making the hub smaller, we must also decrease the thickness of the headstock at that point.--- how much error budget does it eat up if we don't change it? Does the decrease in deflection from the shaft get cancelled out by the increase in deflection due to the thinner headstock?

Choosing Preload Mechanism

Criteria	Wave Spring	Belleville Spring
Height	+	0
Preload ranges	-	+
Availability	+	-

The **free height** (*h*) is the vertical distance between the uppermost and lowermost points on the bottom surface of the spring. It is also the total height of the spring minus the thickness. If the spring is mounted on a solid surface to prevent snap-through, the free height is simply the maximum amount the spring can be deflected. The other geometric variables that govern force/deflection and stress/deflection behavior are the inside diameter (*ID*), outside diameters (*OD*), and the thickness (*t*). For wave springs, the number of waves (*N_w*), and spring width (*b* = *OD*/2-*ID*/2) are also important. In the equations in Table 1 below, contact force is designated as *P*, Maximum Stress as *S*, and deflection as *f*. Poisson's ratio appears as *v*.

	Contact Force (P)		Maximum Stress (S)	
Curved Washer	$P = \frac{4 \cdot E \cdot t^3 \cdot (OD-ID)}{OD^3} \cdot f$		$S = \frac{1.5 \cdot OD}{t^2(OD-ID)} \cdot P = \frac{6 \cdot E \cdot t}{OD^2} \cdot f$	
Wave Spring	$P = \frac{E \cdot b \cdot t^3 \cdot N_w^4}{2.4 \cdot [(OD+ID)/2]^3} \cdot \frac{OD}{ID} \cdot f$		$S = \frac{3 \cdot \pi \cdot [(OD+ID)/2]}{4 \cdot b \cdot t^2 \cdot N_w^2} \cdot P = \frac{3 \cdot \pi \cdot t \cdot N_w^2}{3.2 \cdot [(OD+ID)/2]^2} \cdot \frac{OD}{ID} \cdot f$	
Belleville Washer	$P = \frac{4 \cdot E}{M \cdot (1-v) \cdot (OD)^2} \left[\left(h - \frac{f}{2} \right) \cdot (h-f) \cdot t + t^3 \right] \cdot f$		$S = \frac{4 \cdot E}{M \cdot (1-v) \cdot (OD)^2} \left[C_1 \cdot \left(h - \frac{f}{2} \right) + C_2 \cdot t \right] \cdot f$	
Belleville Washer Constants	$\alpha = \frac{OD}{ID}$	$M = \frac{6}{\pi \cdot \ln \alpha} \left[\frac{(\alpha-1)^2}{\alpha^2} \right]$	$C_1 = \frac{6}{\pi \cdot \ln \alpha} \left[\frac{(\alpha-1)}{\ln \alpha} - 1 \right]$	$C_2 = \frac{6}{\pi \cdot \ln \alpha} \left[\frac{(\alpha-1)}{2} \right]$

Wave Disc Spring

0.76" ID, 1.016" OD, 0.02" Thick



☐ Packs of 10

In stock
\$10.41 per pack of 10
9714K3

[ADD TO ORDER](#)

Spring Type	Disc
Disc Spring Type	Wave
ID	0.76"
ID Tolerance	-0.04" to 0.041"
OD	1.016"
OD Tolerance	-0.05" to 0.051"
Thickness	0.02"
Height	0.067"
Deflection @ Working Load	0.047"
Working Load	101 lbs.
Material	High-Carbon Steel
RoHS	RoHS 3 (2015/863/EU) Compliant
REACH	REACH (EC 1907/2006) (01/16/2020, 205 SVHC) Compliant

Design Trade offs

Bearing Spacing

using headstock thickness=0.6, bearingspacing

=2.6 in

Errum (um)=

0

5.01

25.66

RSS = 26.15

using headstock thickness-0.6,

bearingspaceing=1.6 in

Errum =

0

9.59

23.09

RSS =25.00

small gains by decreasing bearing spacing, but at the cost of bearing life

Overhang/Headstock Thickness Tradeoff

using overhang = 1in, average

headstock thickness=0.6375in

Errum =

0

13.68

26.00

RSS = 29.38

using overhang = 1.225in, average

headstock thickness=.75in

Errum =

0

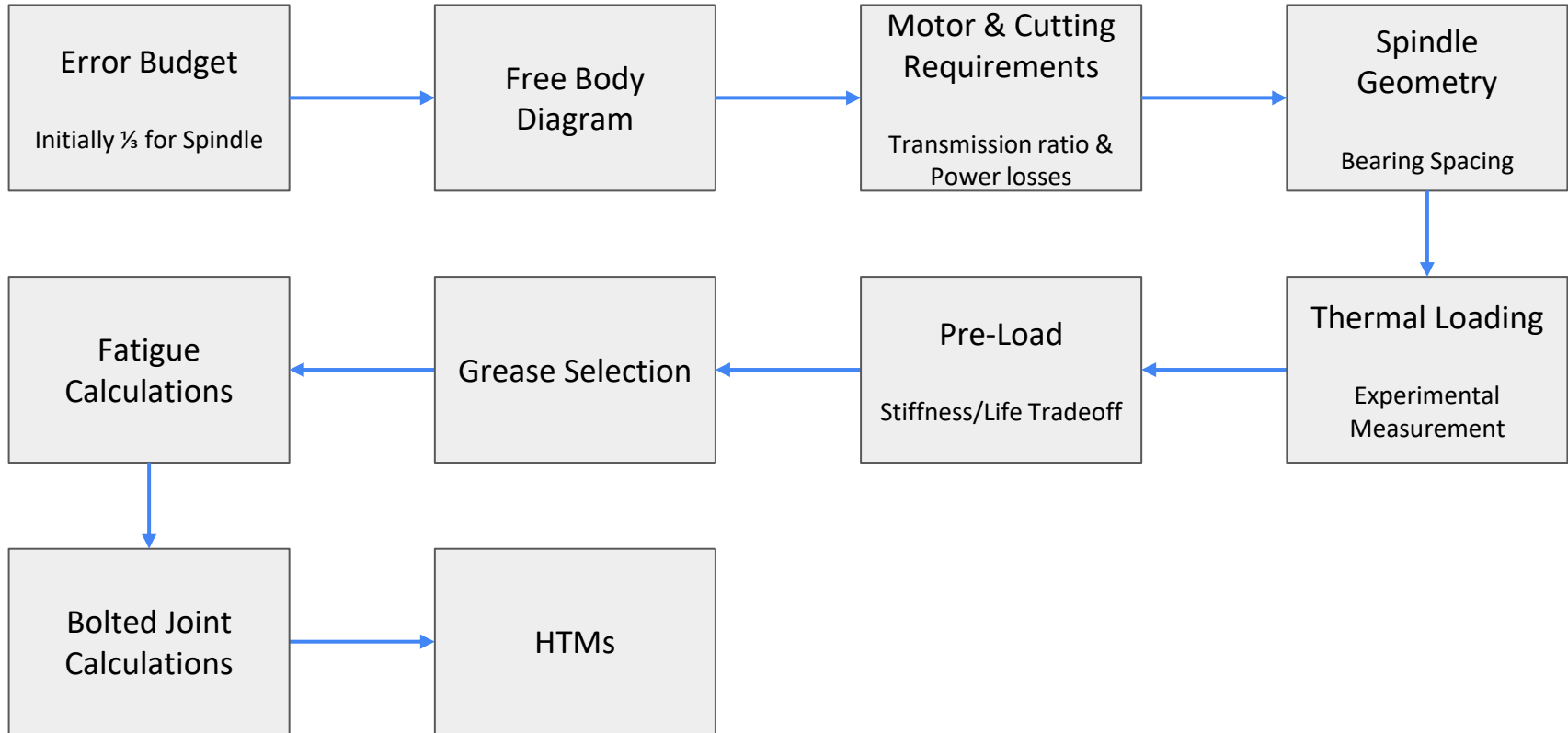
-4.29

16.76

RSS = 17.3

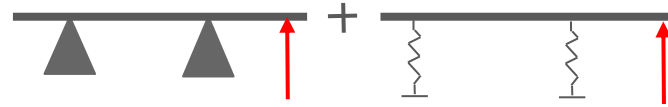
Due to Abbe Errors, thinner headstock portions will allow more deflection.

Design Approach

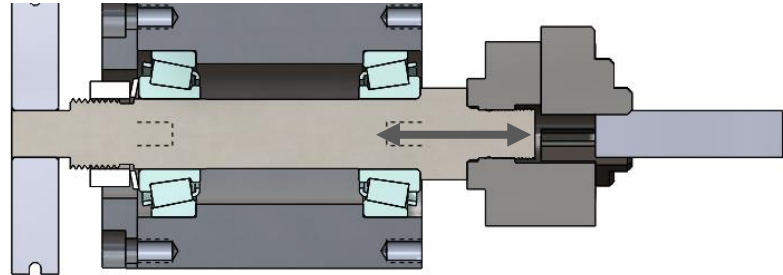


Bearing Preload

$$K_{\text{Spindle_radial}} = \left[\frac{1}{K_{\text{shaft_r}}} + \frac{1}{K_{\text{bearing_r}}} \right]^{-1}$$



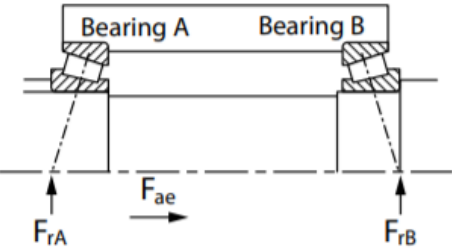
$$K_{\text{Spindle_axial}} = \left[\frac{1}{K_{\text{shaft_a}}} + \frac{1}{K_{\text{bearing_a}}} \right]^{-1}$$



Design Approach (text)

1. Error Budget (from FR) - for starters $\frac{1}{3}$ of total. Rachel working on full model for initial estimates
2. Determine expected loads on shaft/chuck during the turning operation
3. Determine motor and cutting requirements based on FRs and limitations
 - a. Transmission ratio
 - b. Power loss due to bearings, etc.
4. Determine spindle geometry
 - a. Required parameters (chuck, pulley, etc) vs parameters we can vary
 - b. System stiffness : shaft vs bearings vs housing
 - i. Where do we get bang for our buck (bearing spacing)
 - ii. This should show to reduce deflection of shaft
 - iii. Plot of deflection of chuck tip vs lengths we can change.
 - iv. Bearing life should be enough for FR (min)
 - c. Tradeoff between bearing spacing/forces/life
 - d. Justification for all of the parameters we were allowed to vary (hub width, etc) should be for shoulder loading, fillet loading, etc.
5. Thermal Loading considerations
6. Bearing preload selection (washer selection) based on cutting forces
7. Grease selection
8. Spindle shaft life due to fatigue
9. Bolted Joint Considerations?
10. System error budget (stick model T shape- spindle + headstock)
 - a. Show that decreasing HS width to shorten the shaft give us only a few micros less deflection than keeping it thick with slightly longer shaft. Or whatever it turned out to be

TABLE 7. DYNAMIC EQUIVALENT RADIAL LOAD EQUATIONS, SINGLE-ROW TAPERED ROLLER BEARING MOUNTING

Design	Thrust Condition	Axial Load	Dynamic Equivalent Radial Load
	$\frac{0.47 F_{rA}}{K_A} \leq \frac{0.47 F_{rB}}{K_B} + F_{ae}$	$F_{aA} = \frac{0.47 F_{rB}}{K_B} + F_{ae}$ $F_{aB} = \frac{0.47 F_{rB}}{K_B}$	$P_A = 0.4 F_{rA} + K_A F_{aA}^{(1)}$ $P_B = F_{rB}$